

GOTO **COPENHAGEN**

Craft software with confidence in 2026



Engineer Confidence with Reliable Training Programs

Jennifer Petoff, The Reliable PgM™
Co-Editor, *Site Reliability Engineering*





Hello
my name is

Jennifer Petoff
The Reliable PgM™

- Spent 19 years at Google
- Led the SRE EDU team
- Co-editor of the SRE Book (1st & 2nd Edition)
- Ph.D. in Chemistry
- Travel Blogger & Door Photographer at Sidewalk Safari





The Oft Neglected Part of the SDLC



Day 1 - Build

- Write & deploy the code!
- Hopes are high
- Expectations idealized



Day 2 (to ∞) - Run

- Complexity
- System interactions
- Toil!
- Outages!





AI & The Operations Bottleneck

- Build costs are cratering
 - Generative AI has dramatically lowered the effort and time required to write new code
- Operations is (an even bigger) bottleneck
 - Exponential increase in code means corresponding load of reading, verifying, debugging, and maintaining code





Why is training important?





believe in
yourself







Learning → Confidence → Behavior ↔ Culture

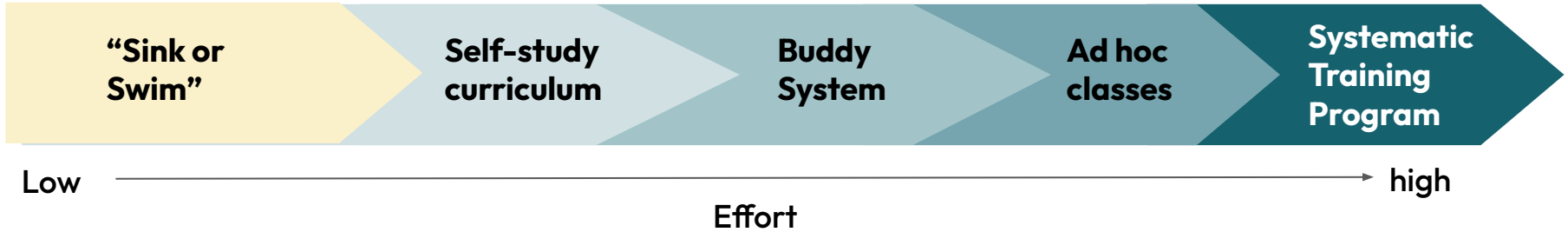


The Reliable PgM™



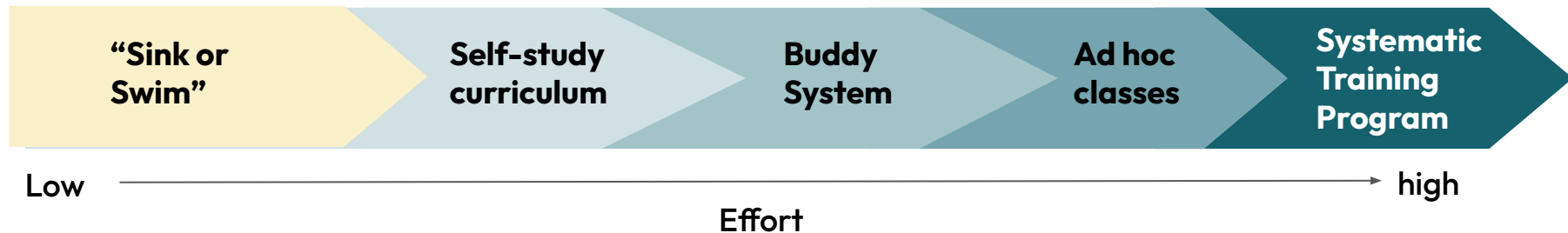


Continuum of Training Options





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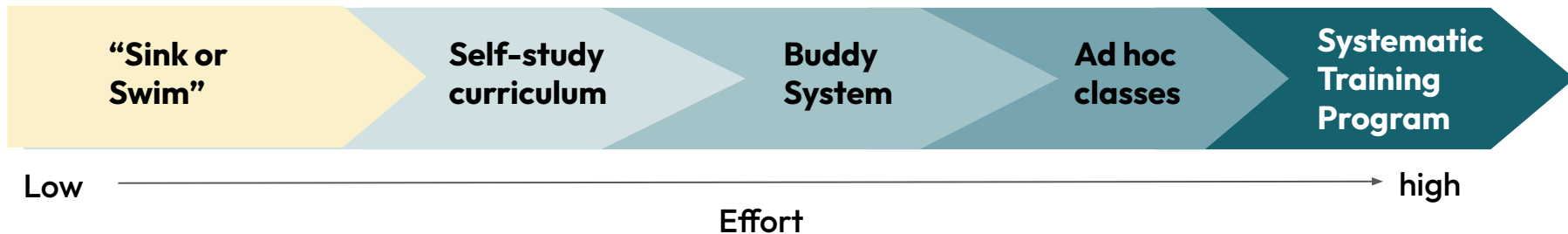
Avoid “Sink or Swim” if you value inclusivity

- Breeds stress, frustration, attrition
- Imposter syndrome





Continuum of Training Options



Higher touch options signal:

- Leadership commitment to the development of employees
- Ensure everyone is speaking with one voice
- Help imbibe the desired culture
- Reinforce desired behaviors



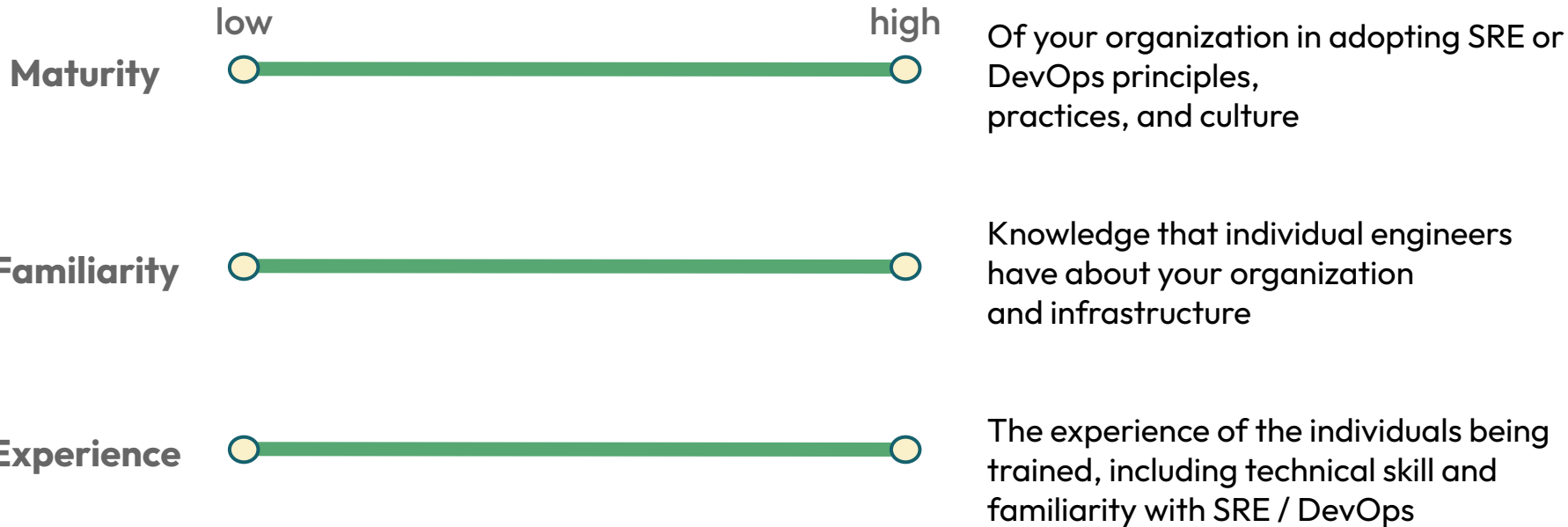


“What should I teach?”





Dimensions That Inform Training Needs







Step 1: Address Any Skill Gaps



- Does your team use a defect tracking system (e.g., Jira, Bugzilla)?
- Does your team have a project planning model (e.g., Agile)?
- Is work done in a version control system (e.g., Git)?
- Are solutions clearly articulated and vetted by team members (e.g., with design docs) before starting implementation?
- Does your team have a shared repository of commonly used tools and libraries?





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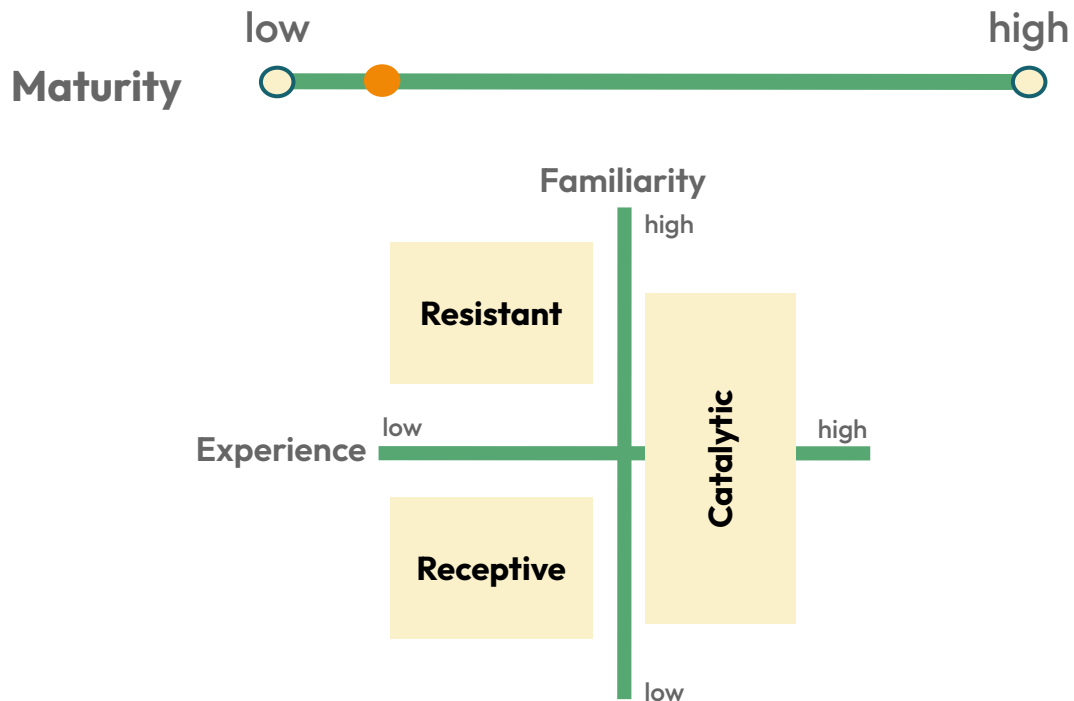


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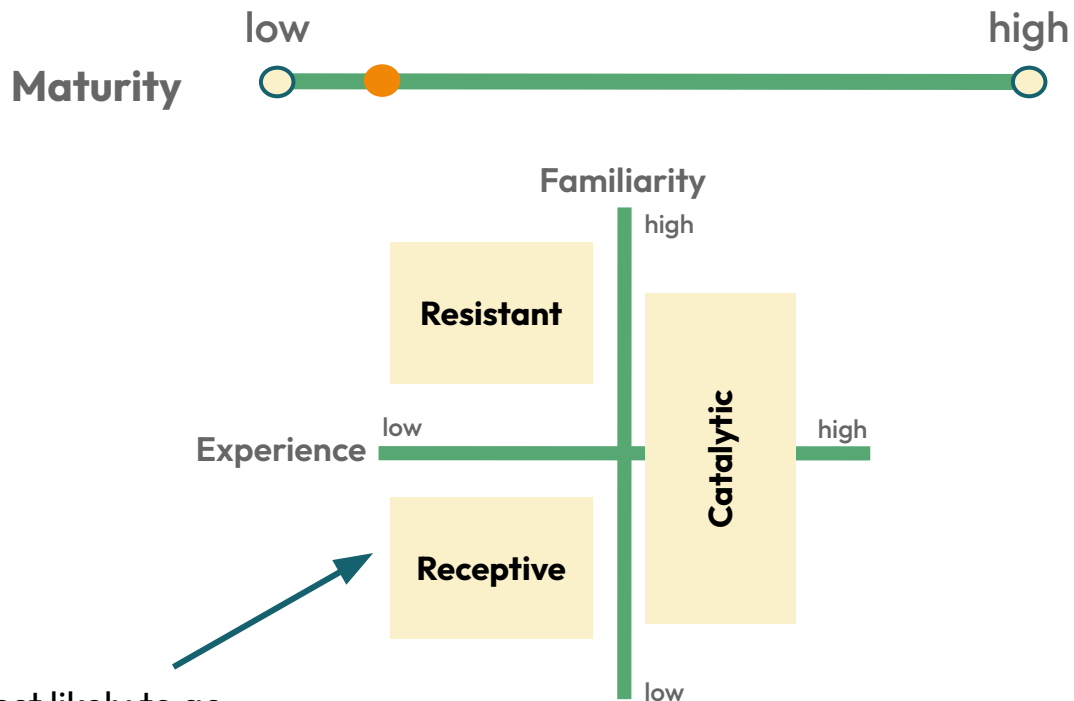


Step 2: Tailor the Message





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Most likely to go
with the flow

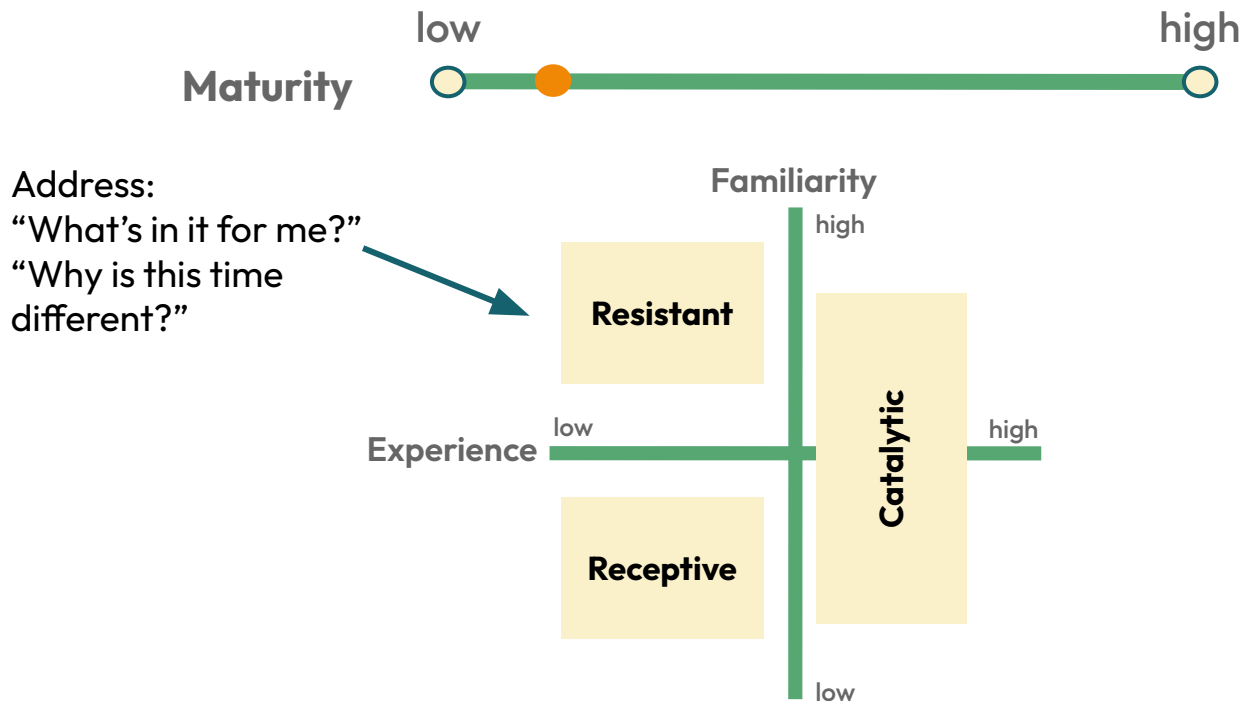


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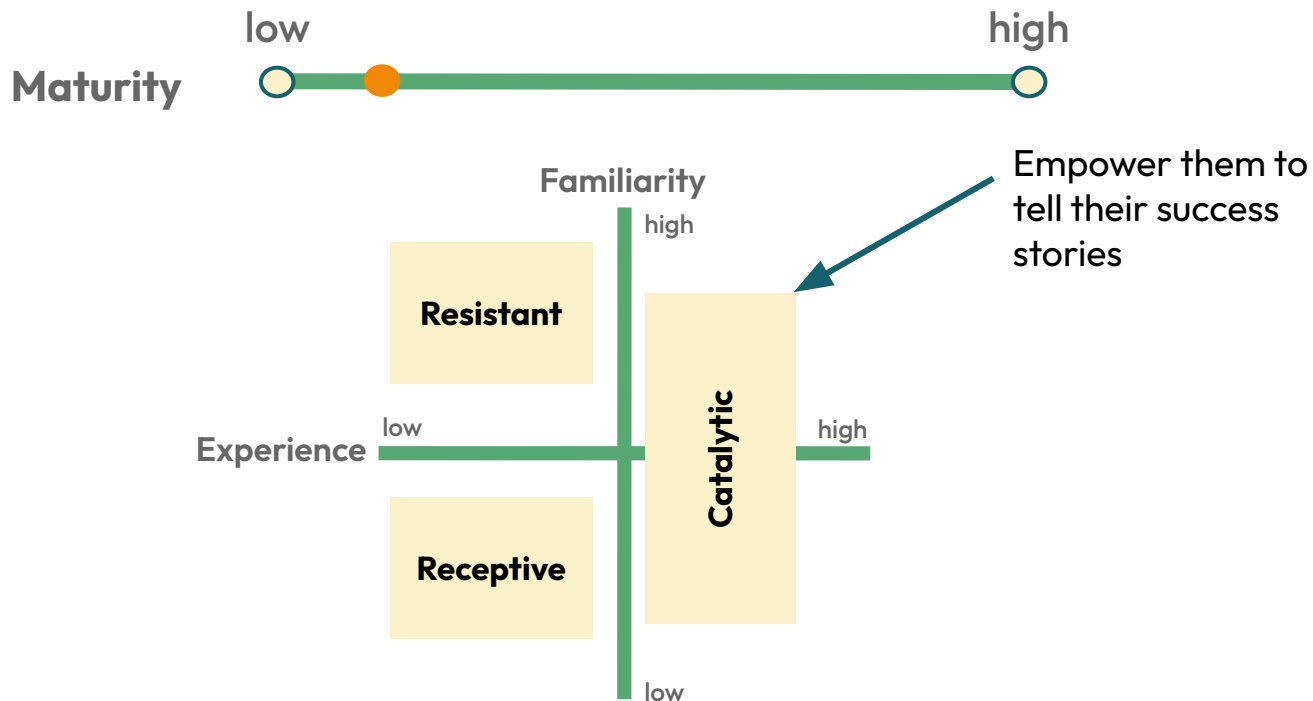


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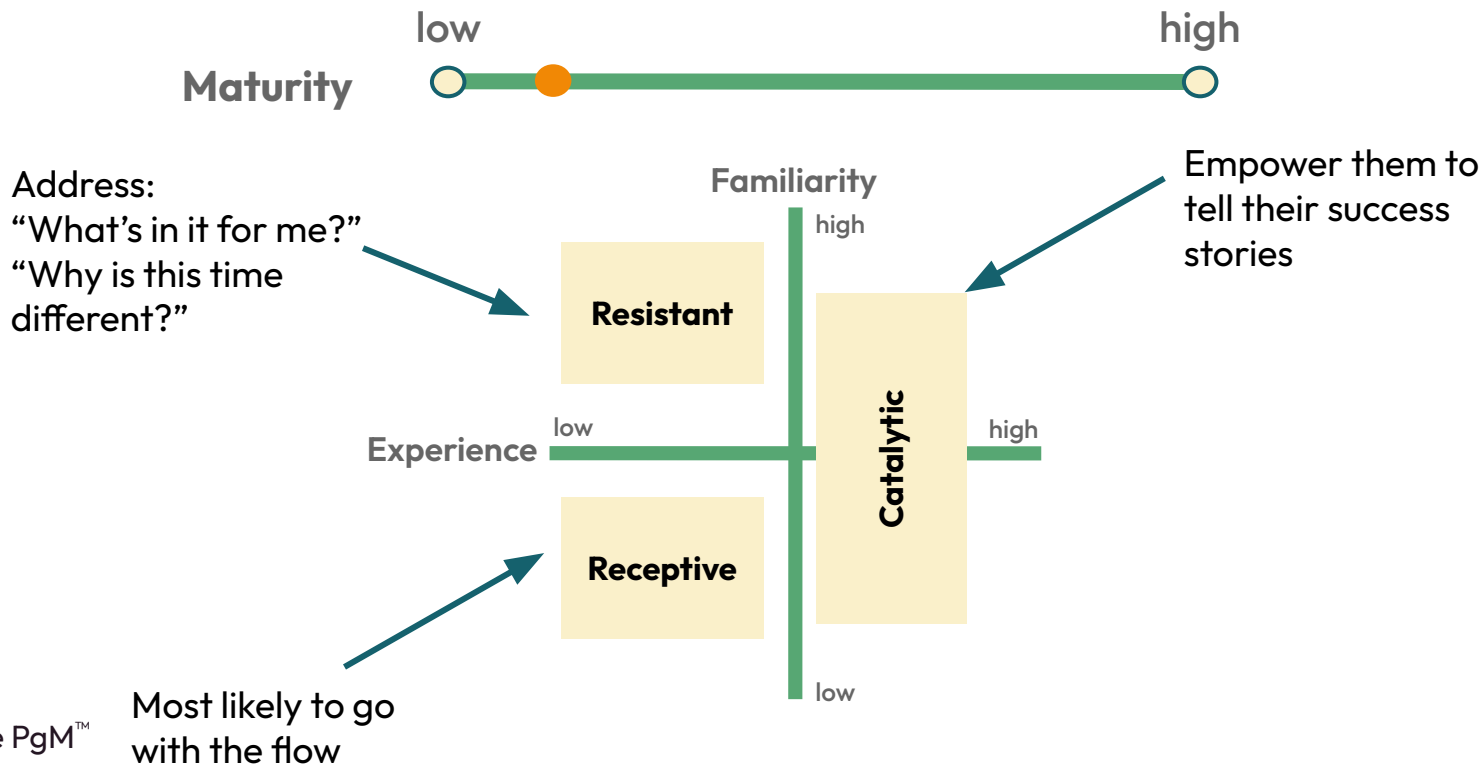


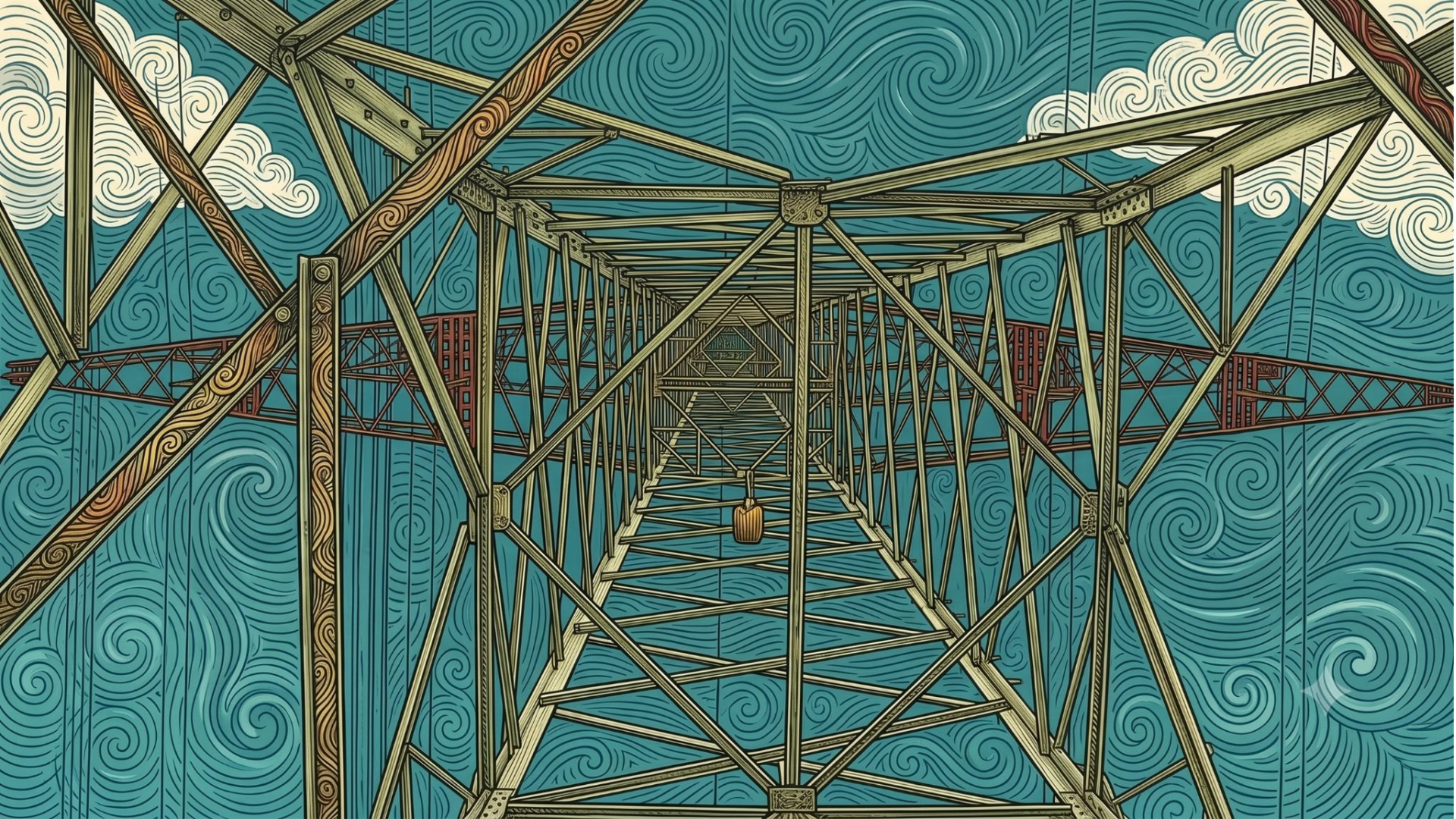
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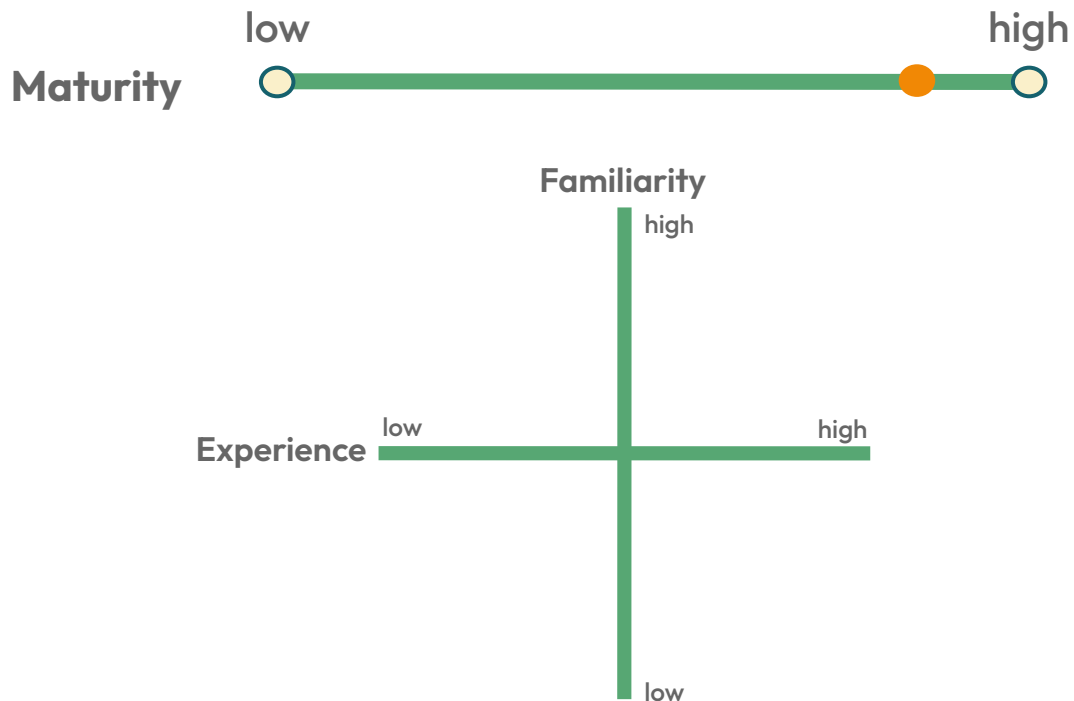
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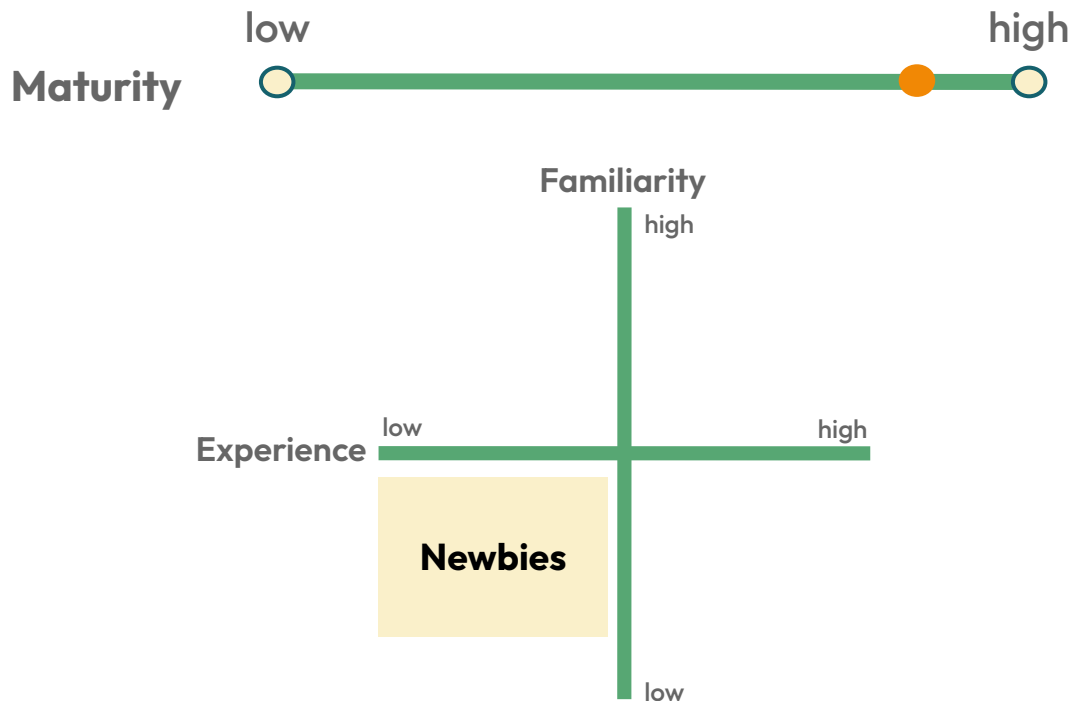


Step 1: Assess the Team Mix



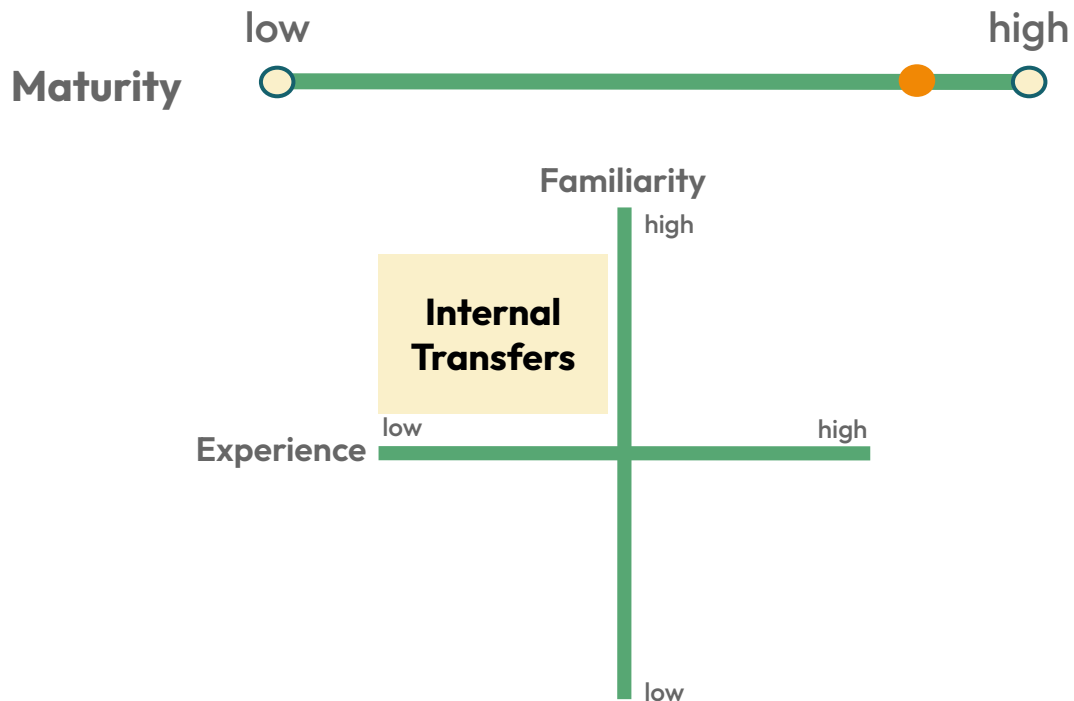


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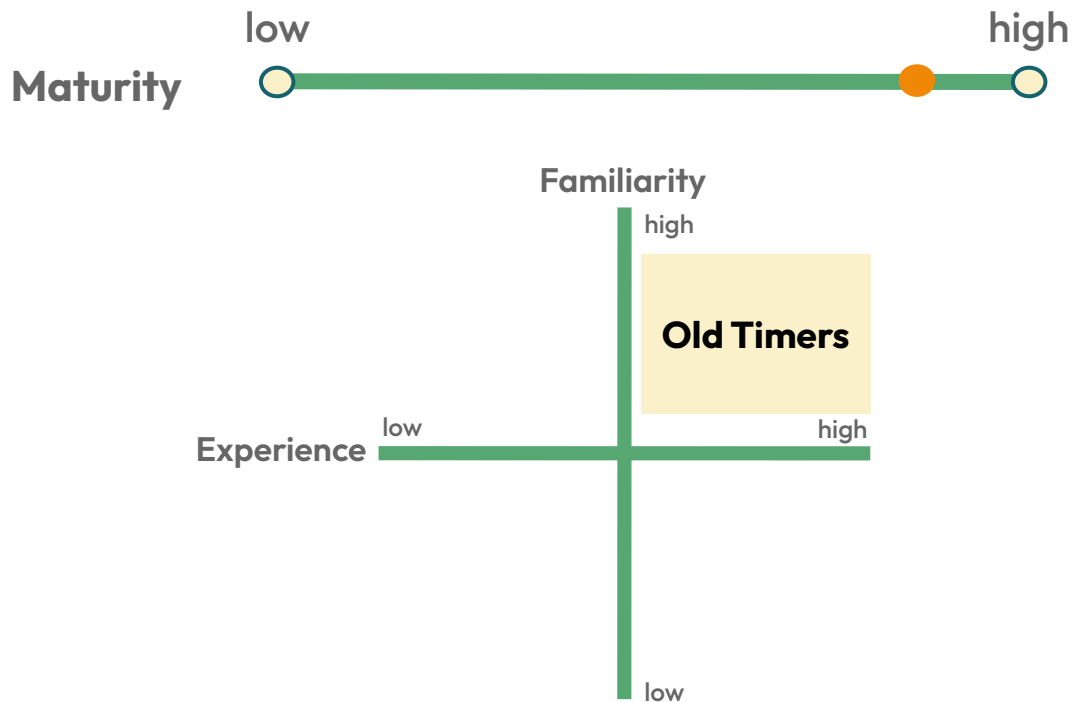


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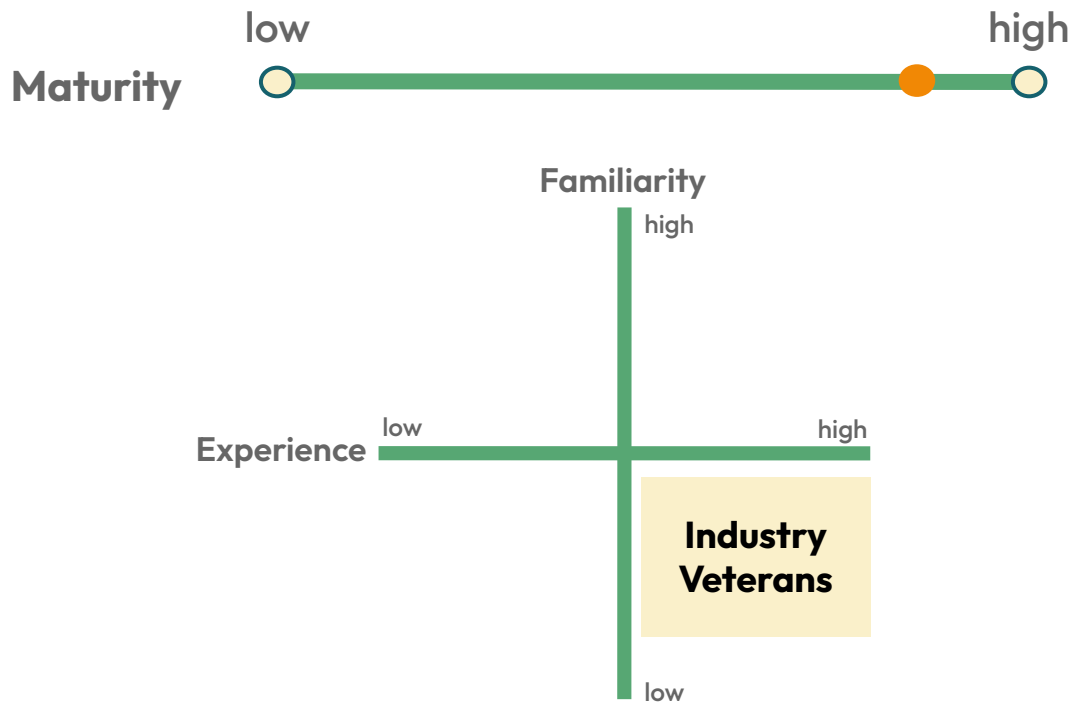


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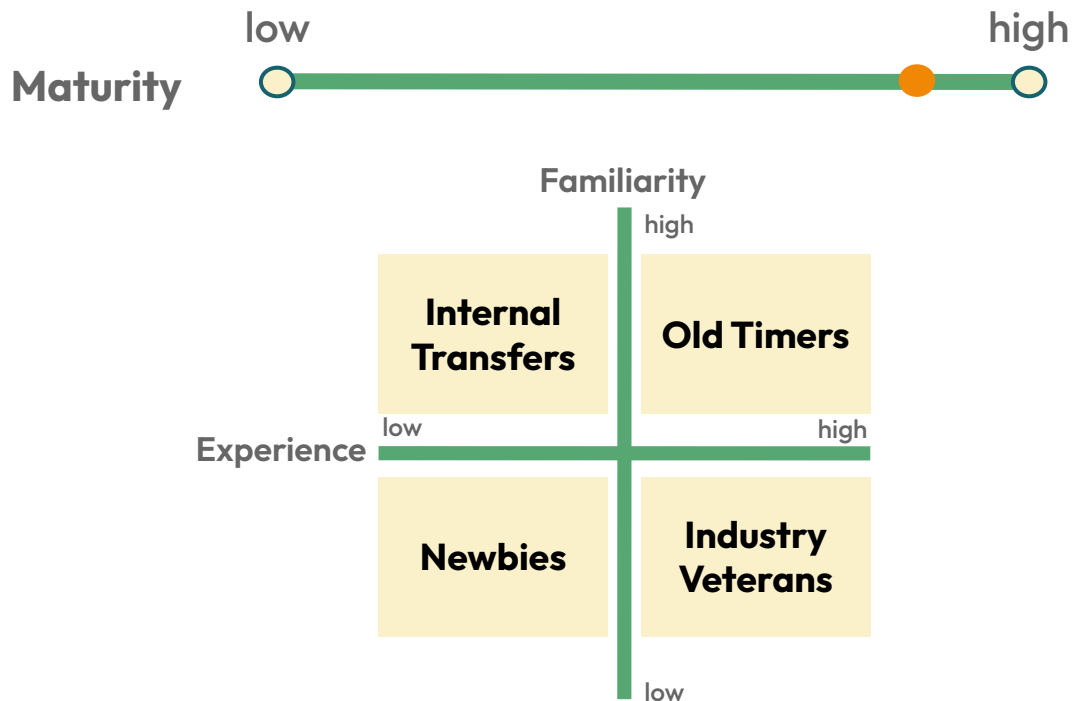


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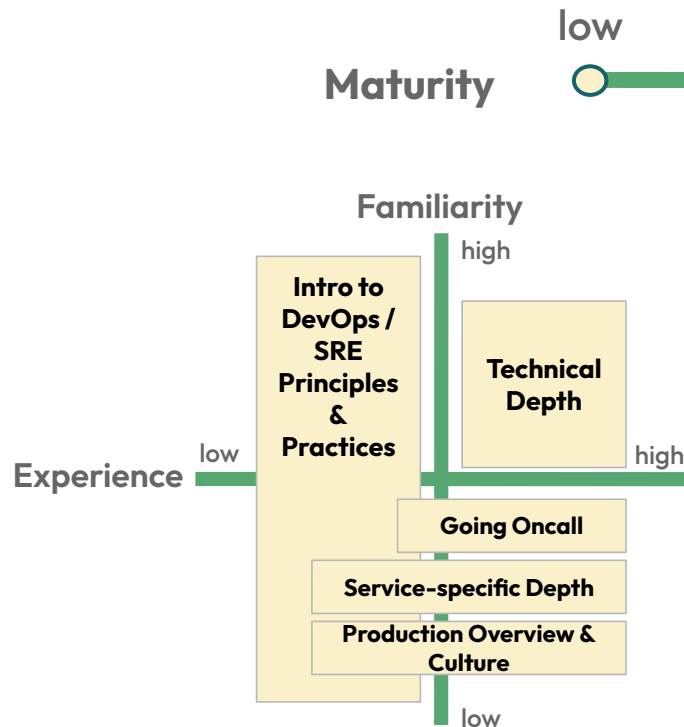


Step 1: Assess the Team Mix





Step 2: Find Your Focus

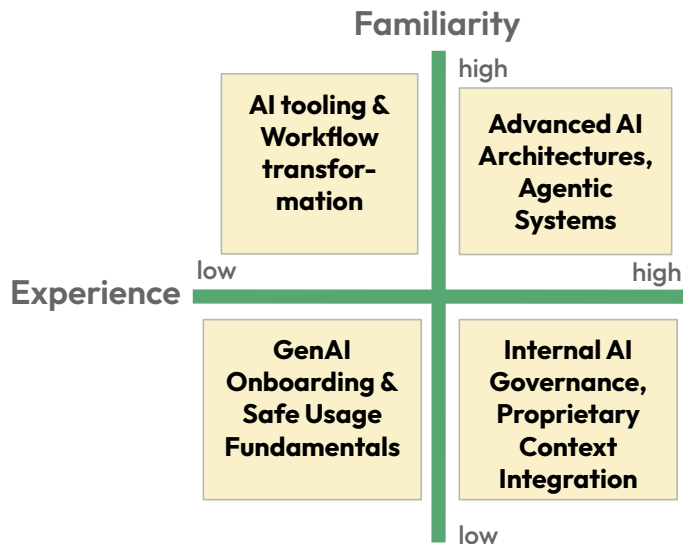


- Newbies need the most attention. They need to learn the infrastructure *and* culture.
- Internal Transfers may know your systems but are less familiar with DevOps/SRE practices so focus here.
- Old Timers are experienced on both dimensions. Go for technical depth to unlock career growth.
- Industry Veterans need to ramp-up on your infrastructure and processes and may need to unlearn some bad habits. Focus on specifics.





This Approach is Generalizable: e.g., AI





WHAT?





The Software Development Analogy

	“What”	“How”
Software Development	Product Features	Deploying to production in a reliable way to meet the needs of our users.
Training Program	Training Content	Deploying a consistent and reliable training program that meets the needs of our students.





How?





Operational Dimensions

Size



How large is your organization?

Growth Rate

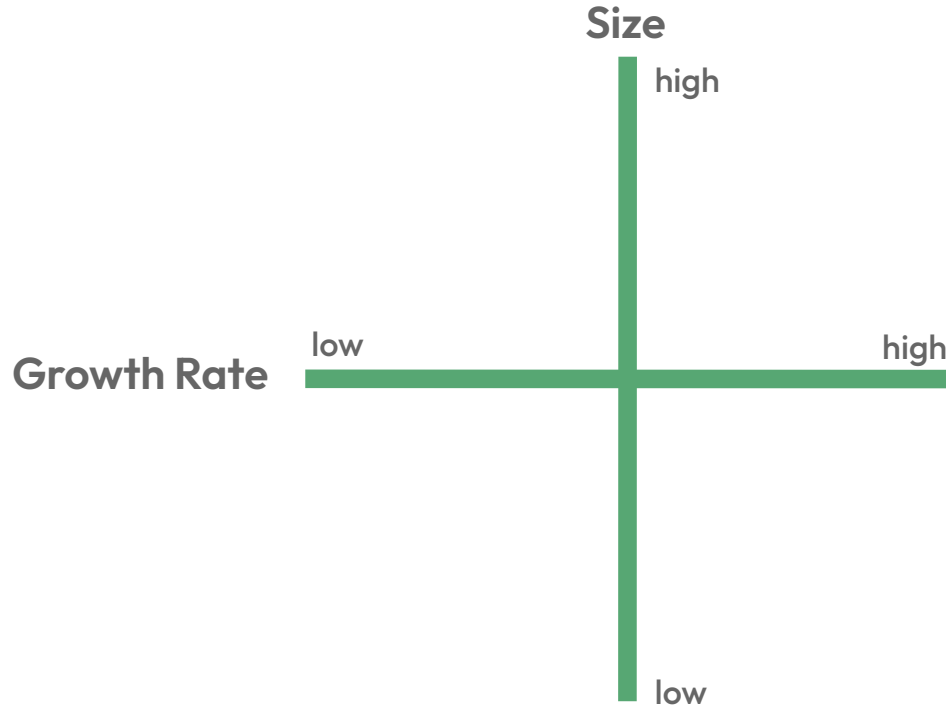


How fast are you growing?



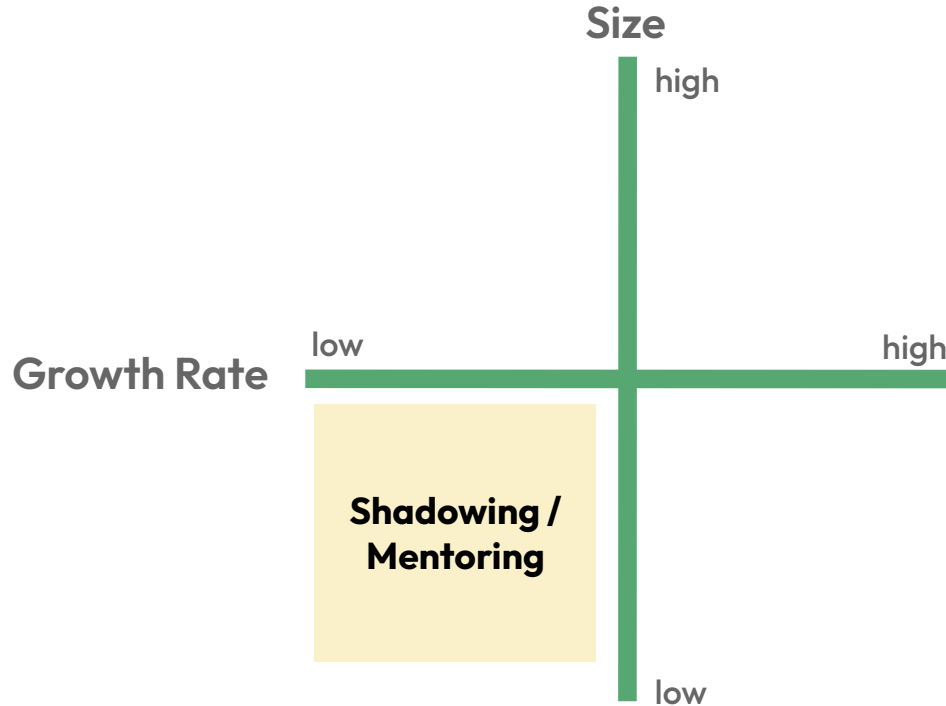


How Much Effort Should You Invest?



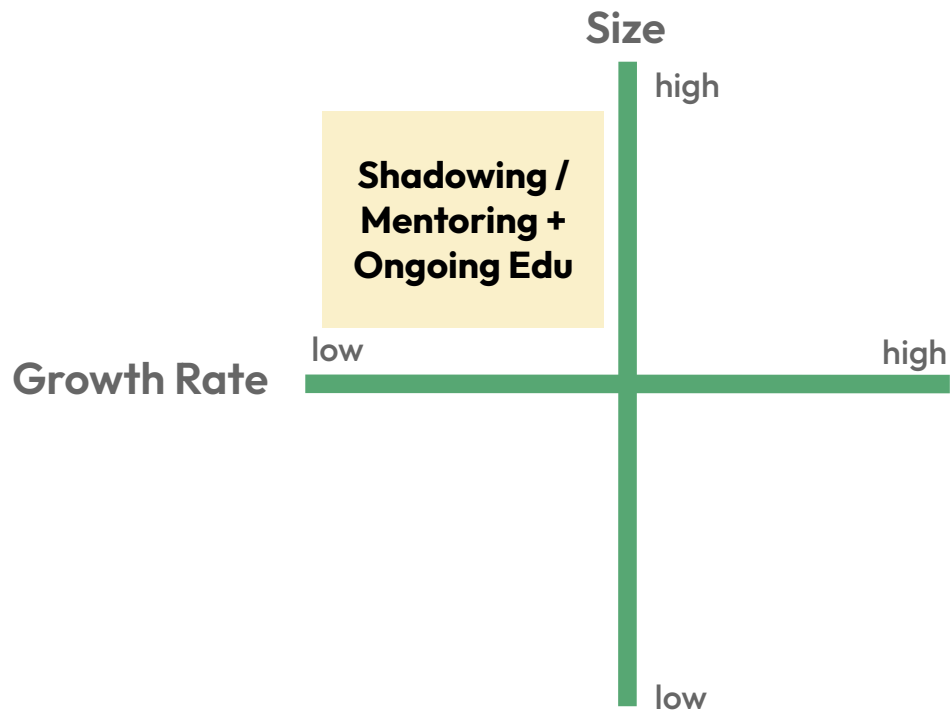


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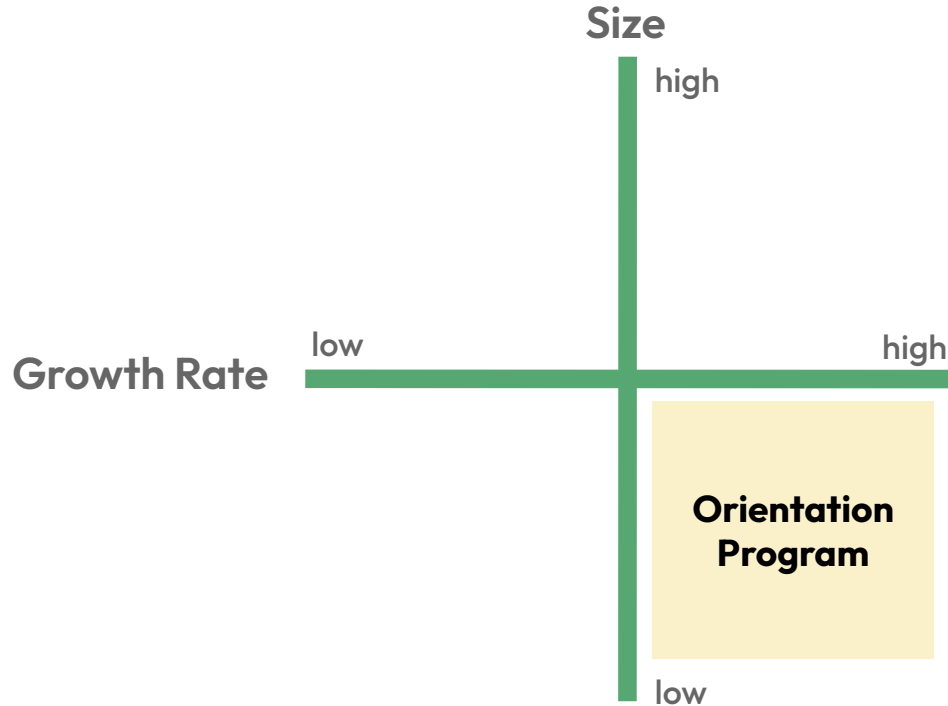


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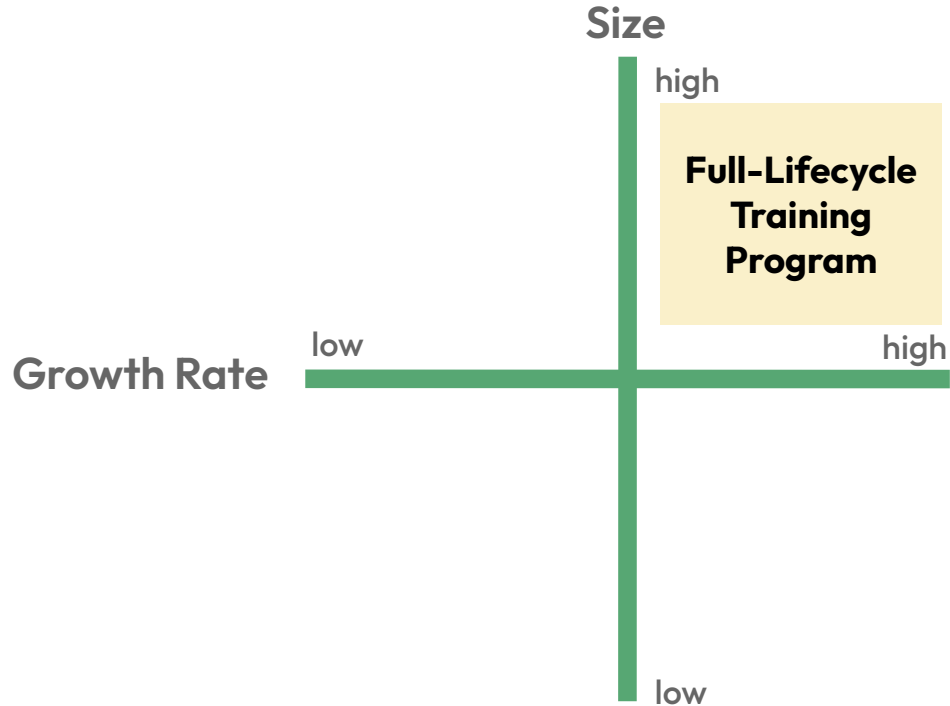


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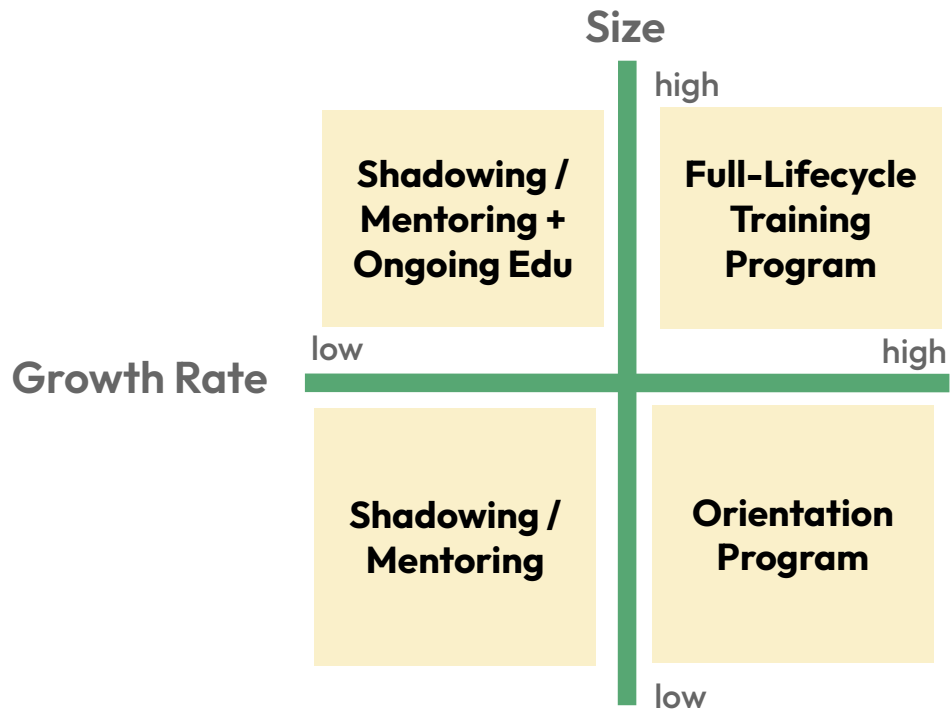


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How Much Effort Should You Invest?



- Invest the most under conditions of rapid growth.
- Small but rapidly growing organizations benefit the most from onboarding training.
- Don't forget about your existing workforce. Invest in ongoing education for their career development.





Why Not Outsource Training to AI?



- Non-deterministic systems can't instill confidence.
- Documentation is not exhaustive
- Organizational lore is passed down person-to-person
- Training fosters connections which lubricate organizational friction.
- Don't underestimate the value of a strong sociotechnical foundation.





Where to Start?



A Student **S**hould **B**e **A**ble **T**o





A Student Should Be Able To

Understand \$foo service?





A Student Should Be Able To

~~Understand \$foo service?~~





A Student Should Be Able To

~~Understand \$foo service?~~

Focus on the behaviors you want to drive:

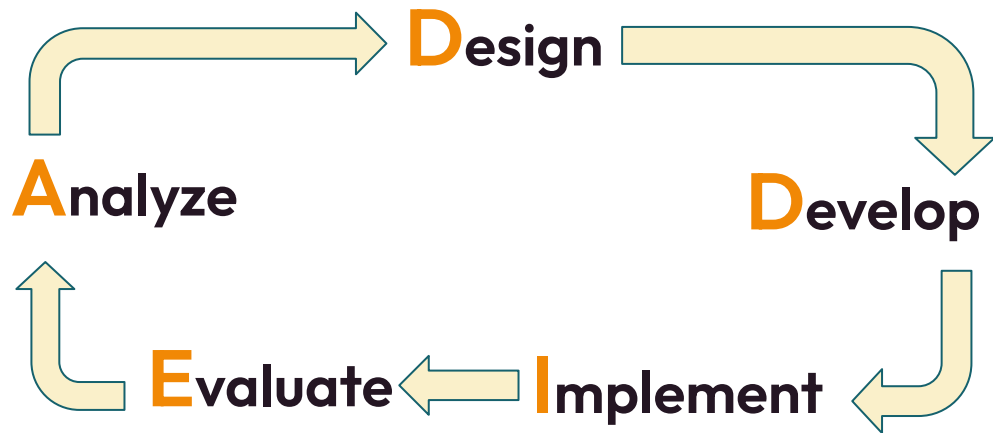
- Use \$tool to identify how much memory a job is using
- Interpret a graph in \$monitoring_tool to identify the health of \$foo service
- Move traffic away from a cluster using \$drain_tool in five minutes

Observe and measure how the training is *applied*





The ADDIE Model of Instructional Design



Analyze needs
Design based on ASSBATS
Develop training content
Implement the program
Evaluate effectiveness

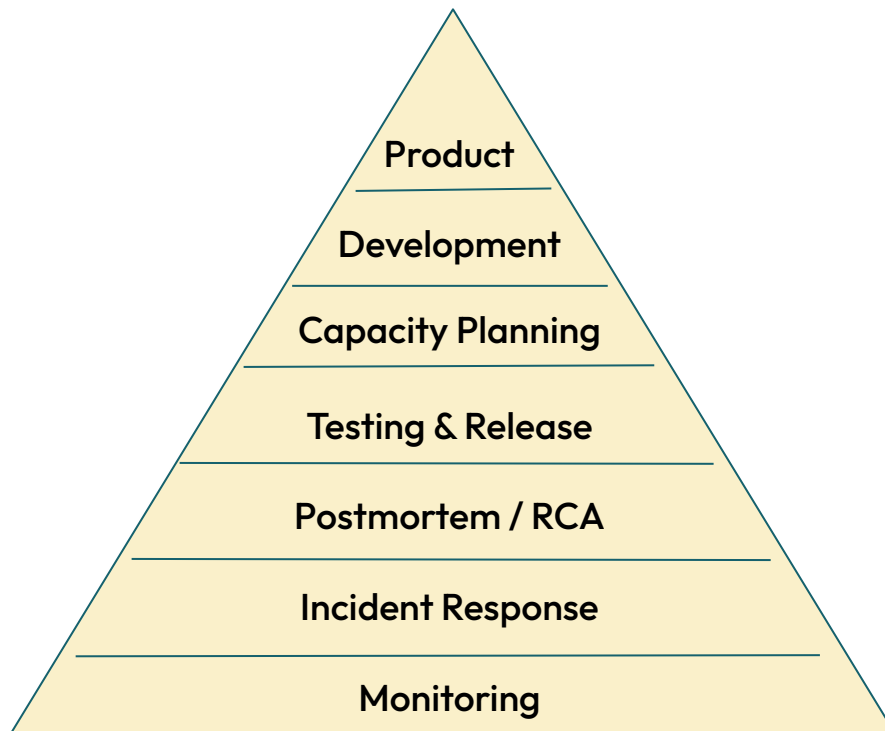


Let's Get Meta for a Minute



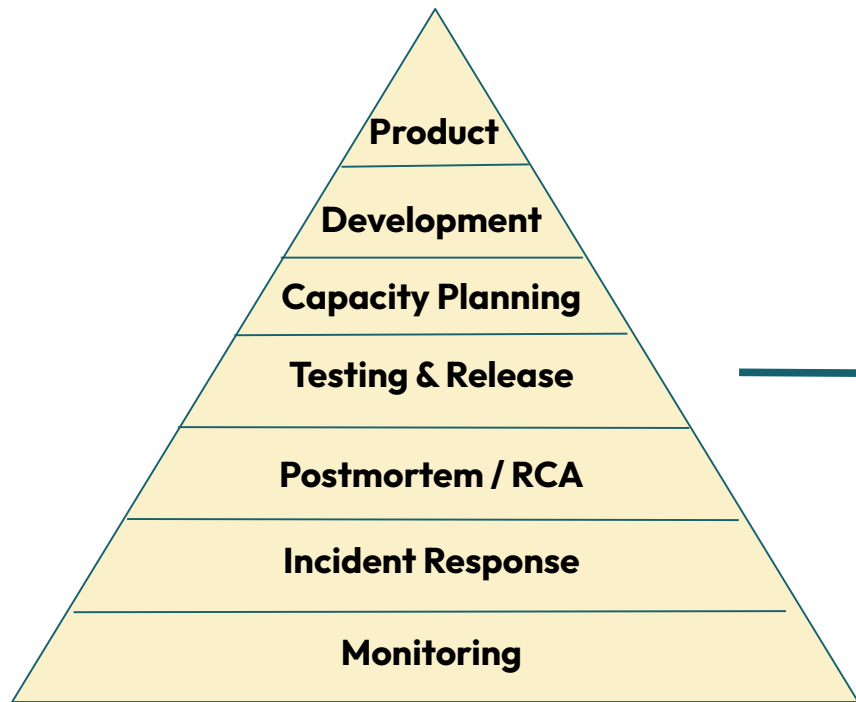


Service Reliability Hierarchy

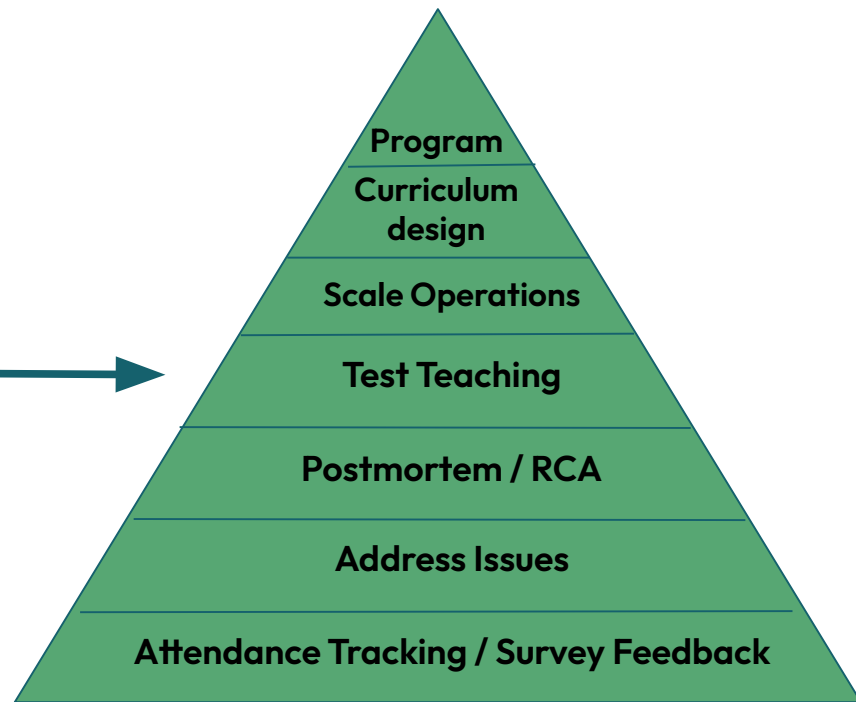




How to SRE a Training Program



Service Reliability Hierarchy*



SRE Training Reliability Hierarchy





Is More Effort Always Better?





Is More Effort Always Better? **No.**





Aim for Good Enough, Not Perfection



SRE Principle in Practice:

- Do just enough to meet the needs of your students.
- Keep them happy, but not too happy.
- Consider trade-offs and *avoid polishing a diamond.*





What Did The Monitoring Say?

Also more prepared, hands-on "Hello world" demonstrations and in-class labs allowing use of the aforementioned paths would be welcome (kinesthetic).

More time doing hands-on work and deeper exploration of how {redacted} were run by SRE teams would be nice.

I disliked the "wall of lecture" in some classes, meaning 1.5 or 2 hours of listening with little/no hands-on exercise.

Some more hands-on activities would have been good.





SRE EDU Orientation v2

- Move away from passive listening
- Troubleshoot a real system, built for this purpose
- Facilitator backs off more and more
- Groups of three students, least experienced in the middle, driving
- Instill confidence



<http://bitly.com/sre-edu>





What Does The Monitoring Say Now?

I went in feeling quite apprehensive & came out feeling like I at least know which way I'm pointed. Thoroughly enjoyed the breakage activities and learning about how Google's infra, monitoring and processes fit together.

Delving into real breaking scenarios was super valuable - I would love more of these (1 per day would be amazing).

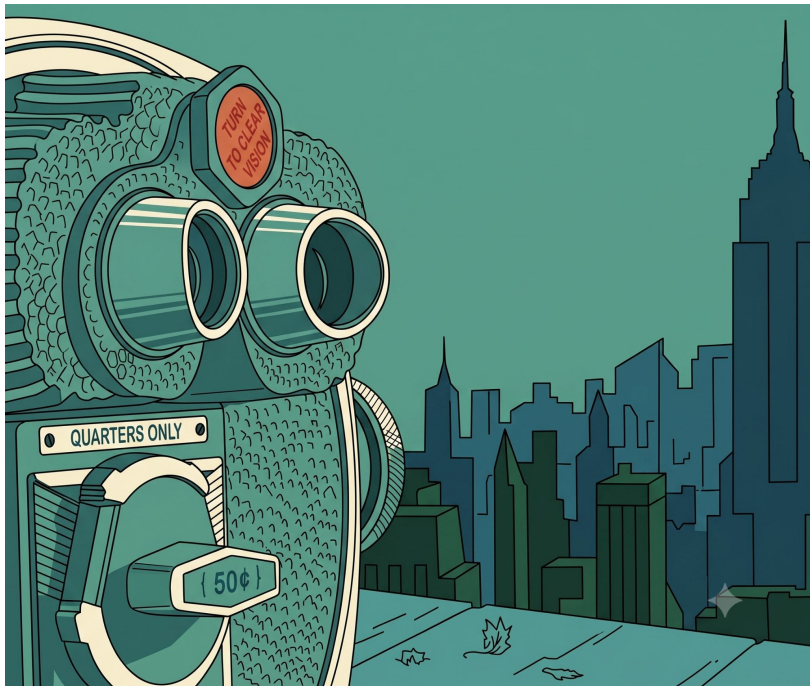
The breakage scenarios in SRE EDU were awesome.

It was the funnest week I've had this year. Overall, it made me feel more connected to production and the technology, which made me really happy.





Better Instrumented for Observability



Concrete behaviors *demonstrated*

- Use a system diagram
- Diagnose issues using SRE tools
- Annotate an outage
- Mitigate a realistic production issue
- Find incident triggers & propose solutions





Scaling SRE EDU Ori v2: Oncall Ready



Throughput Bottleneck: Manual breakage oversight capped at 1,000/yr.

Self-Service Stacks: Automated reservation, provisioning, and environment resets.

Triad Model: Self-organized groups of three with a peer Teaching Assistant.

Runbook Architecture: Step-by-step facilitator playbooks replace central instructors.





Scaling Production Safety to Everyone



Service Model: Treat training infrastructure like production software.

Canary Deployment: Pilot with 1% cohort before global rollout.

Automated Escalations: Automated nudges not manual administrative toil.

Continuous Feedback: Real-time metrics drive rapid curriculum iteration.

Outcomes: Reached 10x+ audience scale, +30 pp NPS increase, >95% completion.





Engineering Confidence... Takeaways





Engineering Confidence... Takeaways

- **A workforce & org culture that is 'production ready'** → non-negotiable in the GenAI era.





Engineering Confidence... Takeaways

- **Learning → Confidence → Behavior → Culture**





Engineering Confidence... Takeaways

- **Training is an investment** → An investment in your organization and people.





Engineering Confidence... Takeaways

- **Evaluate the cost *and* benefits** → to make sure you make the *right* level of investment.





Engineering Confidence... Takeaways

- **Walk the Talk** → Apply SRE / DevOps principles to the training program itself for a consistent and reliable experience.





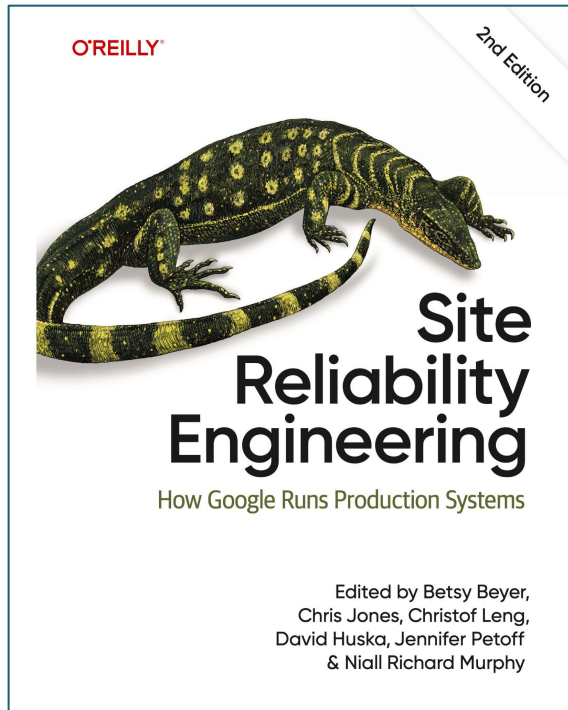
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The SRE Book 2nd Ed. Is (Nearly) Here!

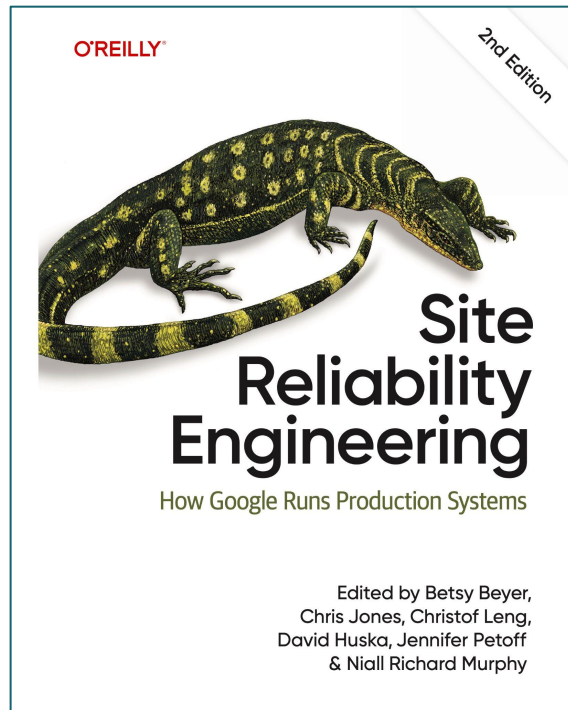




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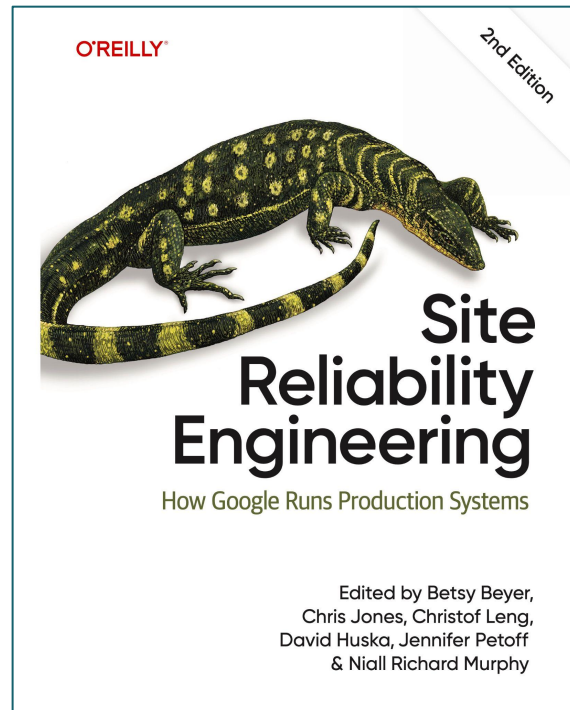
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Origami Might Be

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Thank You!

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